The Journal of Open Source Software

PET2BIDS: a library for converting Positron Emission Tomography data to BIDS

Anthony Galassi [1], **Martin Norgaard** [2,3,4], **Adam G. Thomas** [1], **Robert B. Innis** [1], **Claus Svarer** [2], **Gitte M. Knudsen** [2], **Melanie Ganz** [2,3], **Gabriel Gonzalez-Escamilla** [5], **Cyrus Eierud** [6], **Chris Rorden** [7], **Granville J. Matheson** [8,9], **Murat Bilgel** [10], **and Cyril Pernet** [2]

1 National Institutes of Health, Bethesda, MD, USA **2** Neurobiology Research Unit, Rigshospitalet, Copenhagen, Denmark **3** Department of Computer Science, University of Copenhagen, Copenhagen, Denmark **4** Department of Psychology, Stanford University, CA, USA **5** University Medical Center of the Johannes Gutenberg University Mainz, Mainz, Germany **6** TReNDS Center, Georgia State University, Atlanta, GA, USA **7** Department of Psychology, University of South Carolina, Columbia, SC, USA **8** Mailman school of Public Health, Columbia University, New York, NY, USA **9** Department of Clinical Neuroscience, Karolinska Institutet and Stockholm County Council, Stockholm, Sweden **10** National Institute on Aging Intramural Research Program, Baltimore, MD, USA

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Britta Westner

Reviewers:

- @nbeliy
- @adswa
- @pjtoussaint

Submitted: 04 September 2023
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The Brain Imaging Data Structure (Gorgolewski et al., 2016) is a standard for organizing and naming neuroimaging data, which has quickly become successful and popular in the community with adoption by brain imaging repositories (e.g., OpenNeuro (Markiewicz et al., 2021), PublicnEUro (*Public nEUro*, 2023)), data management tools (e.g., COINS (Landis et al., 2016), XNAT (Marcus et al., 2007)) and computational platforms (e.g. BrainLife (Hayashi S, 2023)). BIDS allows data to be shared much more efficiently, enables the development of automated data analysis pipelines, and improves reproducibility.

The BIDS extension for Positron Emission Tomography (PET-BIDS) (Norgaard et al., 2022) provides nomenclature for structured data and metadata, including all the necessary information to share and report on PET blood and metabolite (Knudsen et al., 2020). Here we present a code library, developed in both Matlab and Python, allowing the conversion of DICOM (*DICOM PS3.3 2020b - Information Object Definitions*, 2020) and ECAT (CTI/Siemens proprietary data format) PET imaging data and metadata (e.g., timing information such as 'time zero' or blood measurements) into files that follow the BIDS specification (nifti, json, tsv).

Statement of need

Conversion tools from native format to BIDS are essential to help researchers to use BIDS. While tools for MRI and MEEG exist, PET2BIDS is the first tool to convert both DICOM and ECAT PET data and PET metadata to BIDS files. Because PET2BIDS is a code library, conversion is done using the command line. PET2BIDS can, however, also be integrated into software (with a graphical user interface) that aims at more general BIDS conversion, and current efforts are underway integrating PET2BIDS with ezBIDS (Levitas et al., 2024) and BIDSCoin (Zwiers et al., 2022). While our library allows to convert and name files, these more generic tools also allow data to be structured following the BIDS specification.

File conversion: The conversion for PET data stored in DICOM format to NIfTI is performed using the `dcm2niix4pet` functions, which are wrappers around dcm2niix (Xiangrui Li et al., 2016; Rorden, 2023). Those functions can extend the image sidecar JSON file generated by

dcm2niix with user-provided metadata, making them BIDS-compliant. The conversion of PET data stored in ECAT format is performed using the dedicated ecat2nii functions. The Matlab code relies on the `readECAT7.m` (Christian & Muzic, 1998) function written by B.T. Christian (1998) and revised by R.F. Muzic (2002) to read the data while writing Nifti files relies on `nii_tool` (X. Li, 2016). The Python code mirrors the Matlab code, except writing of Nifti files, which are delegated to Nibabel (Brett et al., 2023). For both Matlab and Python, the data conversion and writing are thoroughly tested (see ecat validation). Mirroring the DICOM conversion, the ecat2nii functions generate JSON sidecar files from image data, and users must provide additional metadata to make files BIDS-compliant. Note that the Python tools are command line tools, i.e. they can be called directly from a terminal.

PET Metadata: Some radiotracer and pharmaceutical information are always missing in the JSON sidecar files created from reading PET scanner data which is why `dcm2niix4pet` and `ecat2nii` take additional metadata. It is also possible to update existing JSON sidecar files with new metadata directly using `updatejsonpetfile.m` or `update_json_pet_file.py`.

Spreadsheet conversion: Tabular data formats are ubiquitous in the PET community, mainly to keep track of radiotracer information injected per participant and record time and radiotracer concentration from the blood sampling. Functions were created to read preformatted tabular data to create or update existing sidecar JSON files. In addition, dedicated functions were created to convert either a preformatted tabular file or PMOD files (PMOD being a popular commercial pharmacokinetic modeling software (Burger & Buck, 1997)) to a BIDS blood.tsv file with it's sidecar JSON. Supported formats are .xls, .xlsx, .csv, .tsv and .bld.

Acknowledgements

This work was supported by Novo Nordisk fonden (NNF20OC0063277) and the BRAIN initiative (MH002977-01).

References

Brett, M., Markiewicz, C. J., Hanke, M., Côté, M.-A., Cipollini, B., McCarthy, P., Jarecka, D., Cheng, C. P., Halchenko, Y. O., Cottaar, M., Larson, E., Ghosh, S., Wassermann, D., Gerhard, S., Lee, G. R., Baratz, Z., Wang, H.-T., Kastman, E., Kaczmarzyk, J., … freec84. (2023). *Nipy/nibabel: 5.1.0*. Zenodo. https://doi.org/10.5281/zenodo.7795644

Burger, C., & Buck, A. (1997). Requirements and implementation of a flexible kinetic modeling tool. *Journal of Nuclear Medicine*, *38*(11), 1818–1823.

Christian, B. T., & Muzic, R. F. (1998). *readECAT7*.

DICOM PS3.3 2020b - information object definitions. (2020). https://dicom.nema.org/medical/dicom/2020b/output/chtml/part03/PS3.3.html

Gorgolewski, K. J., Auer, T., Calhoun, V. D., Craddock, R. C., Das, S., Duff, E. P., Flandin, G., Ghosh, S. S., Glatard, T., Halchenko, Y. O., Handwerker, D. A., Hanke, M., Keator, D., Li, X., Michael, Z., Maumet, C., Nichols, B. N., Nichols, T. E., Pellman, J., … Poldrack, R. A. (2016). The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. *Scientific Data*, *3*, 160044. https://doi.org/10.1038/sdata.2016.44

Hayashi S, H. A., Caron BA. (2023). Brainlife.io: A decentralized and open source cloud platform to support neuroscience research. *Nature Methods*. https://doi.org/10.1038/s41592-024-02237-2

Knudsen, G. M., Ganz, M., Appelhoff, S., Boellaard, R., Bormans, G., Carson, R. E., Catana, C., Doudet, D., Gee, A. D., Greve, D. N., Gunn, R. N., Halldin, C., Herscovitch, P., Huang,

H., Keller, S. H., Lammertsma, A. A., Lanzenberger, R., Liow, J.-S., Lohith, T. G., … Innis, R. B. (2020). Guidelines for the content and format of PET brain data in publications and archives: A consensus paper. *Journal of Cerebral Blood Flow & Metabolism*, *40*(8), 1576–1585. https://doi.org/10.1177/0271678X20905433

Landis, D., Courtney, W., Dieringer, C., Kelly, R., King, M., Miller, B., Wang, R., Wood, D., Turner, J. A., & Calhoun, V. D. (2016). COINS data exchange: An open platform for compiling, curating, and disseminating neuroimaging data. *NeuroImage*, *124*, 1084–1088. https://doi.org/10.1016/j.neuroimage.2015.05.049

Levitas, D., Hayashi, S., Vinci-Booher, S., Heinsfeld, Anibal, Bhatia, D., Lee, N., Galassi, A., Niso, G., & Pestilli, F. (2024). ezBIDS: Guided standardization of neuroimaging data interoperable with major data archives and platforms. *Scientifc Data*, *11*. https://doi.org/10.1038/s41597-024-02959-0

Li, X. (2016). *dicm2nii*. https://github.com/xiangruili/dicm2nii

Li, Xiangrui, Morgan, P. S., Ashburner, J., Smith, J., & Rorden, C. (2016). The first step for neuroimaging data analysis: DICOM to NIfTI conversion. *Journal of Neuroscience Methods*, *264*, 47–56. https://doi.org/10.1016/j.jneumeth.2016.03.001

Marcus, D. S., Olsen, T. R., Ramaratnam, M., & Buckner, R. L. (2007). The extensible neuroimaging archive toolkit: An informatics platform for managing, exploring, and sharing neuroimaging data. *Neuroinformatics*, *5*(1), 11–34. https://doi.org/10.1385/ni:5:1:11

Markiewicz, C. J., Gorgolewski, K. J., Feingold, F., Blair, R., Halchenko, Y. O., Miller, E., Hardcastle, N., Wexler, J., Esteban, O., Goncavles, M., Jwa, A., & Poldrack, R. (2021). OpenNeuro. *eLife*. https://doi.org/10.7554/eLife.71774

Norgaard, M., Matheson, G. J., Hansen, H. D., Thomas, A., Searle, G., Rizzo, G., Veronese, M., Giacomel, A., Yaqub, M., Tonietto, M., Funck, T., Gillman, A., Boniface, H., Routier, A., Dalenberg, J. R., Betthauser, T., Feingold, F., Markiewicz, C. J., Gorgolewski, K. J., … Ganz, M. (2022). PET-BIDS, an extension to the brain imaging data structure for positron emission tomography. *Scientific Data*, *9*(1), 65. https://doi.org/10.1038/s41597-022-01164-1

Public nEUro. (2023). https://public-neuro.github.io/

Rorden, C. (2023). *dcm2nii*. http://www.mccauslandcenter.sc.edu/mricro/mricron/dcm2nii.html

Zwiers, M. P., Moia, S., & Oostenveld, R. (2022). BIDScoin: A user-friendly application to convert source data to brain imaging data structure. *Frontiers in Neuroinformatics*, *15*. https://doi.org/10.3389/fninf.2021.770608

DRAFT